Hidden-Bits Generators with Succinct CRS and Openings

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Zero-Knowledge Proofs



Prover

Knows a witness w

Public statement: $x \in \mathscr{L}$



Verifier

Zero-Knowledge Proofs



Public statement: $x \in \mathscr{L}$

Proof protocol



Prover

Knows a witness w

Verifier

Checks the proof

Zero-Knowledge Proofs



Public statement: $x \in \mathscr{L}$

Proof protocol





Prover

Knows a witness w

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Completeness An honest prover convinces the verifier of a true statement.

Zero-Knowledge Proofs



Public statement: $x \in \mathscr{L}$

Proof protocol





Prover

Knows a witness w

Verifier

Checks the proof

Completeness	An honest prover convinces the verifier of a true statement.
Statistical Soundness	A cheating prover can convince the verifier of a false statement only with tiny probability.

Zero-Knowledge Proofs



Public statement: $x \in \mathscr{L}$

Proof protocol





Prover

Knows a witness w

Verifier

Checks the proof

Completeness	An honest prover convinces the verifier of a true statement.
Statistical Soundness	A cheating prover can convince the verifier of a false statement only with tiny probability.
Zero knowledge	The verifier learns nothing beyond the statement being true.

Non-Interactive Zero-Knowledge Proofs

The prover sends just one message.

Public statement: $x \in \mathscr{L}$

crs

Proof π



Prover

Knows a witness w



Verifier

Checks the proof

NIZKs from Hidden-Bits Generators

Hidden-Bits Generator



NIZK proofs

Hidden-Bits Generators

Generate k pseudorandom bits with a short commitment.

Public setup: crs



Prover



Verifier

Hidden-Bits Generators

Generate k pseudorandom bits with a short commitment.



Prover

Knows $\mathbf{b} \in \{0,1\}^k$

Public setup: crs

Short commitment com



Verifier

Hidden-Bits Generators

Generate k pseudorandom bits with a short commitment.



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Knows $\mathbf{b} \in \{0,1\}^k$

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Open S with $\mathbf{b}_S$ and π



Verifier

Checks openings

Hidden-Bits Generators

Generate k pseudorandom bits with a short commitment.



Public setup: crs

Short commitment com

Open S with $\mathbf{b}_S$ and π



Prover

Knows $\mathbf{b} \in \{0,1\}^k$

Verifier

Checks openings

Statistical soundness com statistically binds to $\mathbf{b}$

Hidden-Bits Generators

Generate k pseudorandom bits with a short commitment.



Public setup: crs

Short commitment com

Open S with $\mathbf{b}_S$ and π



Prover

Knows $\mathbf{b} \in \{0,1\}^k$

Verifier

Checks openings

Statistical soundness com statistically binds to $\mathbf{b}$

Hiding Unopened bits remain pseudorandom, given $\mathbf{b}_S$ and π.

HBG with Short CRS or Openings

Question

How small can the CRS and openings be in HBG?

Our Results

LWE

CRS size: $\mathrm{poly}(\lambda, \log k)$

Previously: $\geq k \cdot \mathrm{poly}(\lambda)$
[Wat24,WWW25,BCDJMS25]

λ: security parameter. k: hidden bits. Opening bounds exclude revealed bit values.

Our Results

LWE

CRS size: $\mathsf{poly}(\lambda, \log k)$

Previously: $\geq k \cdot \mathsf{poly}(\lambda)$
[Wat24,WWW25,BCDJMS25]

Pairings

Opening size: $\mathsf{poly}(\lambda)$

NIZK proofs:
$\tilde{\mathcal{O}}(|C|) + \mathsf{poly}(\lambda, \log |C|)$

λ: security parameter. k: hidden bits. Opening bounds exclude revealed bit values.

Our Results

LWE

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Pairings

Opening size: $\mathsf{poly}(\lambda)$

NIZK proofs:
$\tilde{\mathcal{O}}(|C|) + \mathsf{poly}(\lambda, \log |C|)$

RSA

CRS and opening size:
$\mathsf{poly}(\lambda)$

NIZK proofs:
$\tilde{\mathcal{O}}(|C|) + \mathsf{poly}(\lambda, \log |C|)$

λ: security parameter. k: hidden bits. Opening bounds exclude revealed bit values.

Our Results

LWE

CRS size: poly$(\lambda, \log k)$

Previously: $\geq k \cdot \text{poly}(\lambda)$
[Wat24,WWW25,BCDJMS25]

Pairings

Opening size: poly(λ)

NIZK proofs:
$\tilde{\mathcal{O}}(C) + \text{poly}(\lambda, |C|)$

RSA

CRS and opening size:
poly(λ)

NIZK proofs:
$\tilde{\mathcal{O}}(C) + \text{poly}(\lambda, |C|)$

λ: security parameter. k: hidden bits. Opening bounds exclude revealed bit values.

HBG from LWE - CRS and Commitment

CRS

$$\mathsf{crs} = (\mathbf{A}, \mathbf{B_0}, \mathbf{B_1}, \mathbf{C_0}, \mathbf{C}_1, \mathbf{V}_0, \mathbf{V}_1)$$

$$\mathbf{B}_i \in \{0,1\}^{k \times m}$$

$$\mathbf{C}_i = \mathbf{B}_i(\mathbf{SA} + \mathbf{E}) + \mathbf{E} + \mathbf{TG}_i$$

$$\mathbf{V}_i = \mathbf{B}_i(\mathbf{TA} + \mathbf{F}) + \mathbf{F}$$

Inspired by [AMR25]

HBG from LWE - CRS and Commitment

CRS

$$\text{crs} = (\mathbf{A}, \mathbf{B_0}, \mathbf{B_1}, \mathbf{C_0}, \mathbf{C}_1, \mathbf{V}_0, \mathbf{V}_1)$$

$$\mathbf{B}_i \in \{0,1\}^{k \times m}$$

$$\mathbf{C}_i = \mathbf{B}_i(\mathbf{SA} + \mathbf{E}) + \mathbf{E} + \mathbf{TG}_i$$

$$\mathbf{V}_i = \mathbf{B}_i(\mathbf{TA} + \mathbf{F}) + \mathbf{F}$$

Commitment

$$\text{com} = \mathbf{h} = \mathbf{A} \cdot \pi$$

$$\mathbf{h}_0 = \mathbf{A} \cdot \mathbf{x}_0 \qquad \mathbf{h}_1 = \mathbf{A} \cdot \mathbf{x}_1$$

$$\pi = \begin{pmatrix} \mathbf{G}^{-1}(\mathbf{h}_0) \\ \mathbf{G}^{-1}(\mathbf{h}_1) \end{pmatrix}$$

Inspired by [AMR25]

HBG from LWE - Open and Statistical Sound

GenBit 1st coord.

Root to leaf path $(\pi, \mathbf{x}_0)$

$$\mathbf{y}_0 = \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0$$

$$\alpha_0 = \mathsf{Round}(\mathbf{y}_0)$$

HBG from LWE - Open and Statistical Sound

GenBit 1st coord.

Root to leaf path $(\pi, \mathbf{x}_0)$

$$\mathbf{y}_0 = \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0$$

$$\alpha_0 = \text{Round}(\mathbf{y}_0)$$

Open

Root-to-leaf path

HBG from LWE - Open and Statistical Sound

GenBit 1st coord. | **Open**

Root to leaf path $(\pi, \mathbf{x}_0)$

$$\mathbf{y}_0 = \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0$$

$$\alpha_0 = \mathrm{Round}(\mathbf{y}_0)$$

Root-to-leaf path

$$\begin{aligned}\mathbf{y}_0 &= \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0 \\ &= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\underbrace{\mathbf{A}\pi}_{=\mathbf{h}} + \mathbf{B}_0\mathbf{T}\underbrace{(\mathbf{G}_0\pi - \mathbf{A}\mathbf{x}_0)}_{=0} + \mathbf{e}\end{aligned}$$

$$= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\mathbf{h} + \mathbf{e}$$

HBG from LWE - Open and Statistical Sound

GenBit 1st coord. | **Open** | **Trapdoor**

Root to leaf path $(\pi, \mathbf{x}_0)$

$\mathbf{y}_0 = \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0$

$\alpha_0 = \text{Round}(\mathbf{y}_0)$

Root-to-leaf path

$\alpha_0 = \text{Round}(\mathbf{B}_0\mathbf{B}_0\mathbf{S}\mathbf{h})$

$$\begin{aligned}\mathbf{y}_0 &= \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0 \\ &= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\underbrace{\mathbf{A}\pi}_{=\mathbf{h}} + \mathbf{B}_0\mathbf{T}\underbrace{(\mathbf{G}_0\pi - \mathbf{A}\mathbf{x}_0)}_{=0} + \mathbf{e} \\ &= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\mathbf{h} + \mathbf{e}\end{aligned}$$

HBG from LWE - Open and Statistical Sound

Statistical Soundness

Given $\mathbf{h}$ and td we can compute α_0

$$\mathbf{y}_0 = \mathbf{B}_0\mathbf{C}_0\pi - \mathbf{V}_0\mathbf{x}_0$$

$$= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\underbrace{\mathbf{A}\pi}_{=\mathbf{h}} + \mathbf{B}_0\mathbf{T}\underbrace{(\mathbf{G}_0\pi - \mathbf{A}\mathbf{x}_0)}_{=0} + \mathbf{e}$$

$$= \mathbf{B}_0\mathbf{B}_0\mathbf{S}\mathbf{h} + \mathbf{e}$$

Missing Details

- Tree via recursion - crs of size $\mathsf{poly}(\lambda, \log k)$
- Hiding: Uniform crs via LWE + LHL
- Polynomial modulus-to-noise ratio via gadget matrices

Recap

Results: New HBG

- LWE: short CRS
- Pairings: short proof
- RSA: short proof and CRS

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- LWE: short CRS
- Pairings: short proof
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Open problems:

- LWE: short proofs
- Pairings: short CRS